Strategy Research Report

Systematic evaluation of a chart-analysis note
against 25+ years of daily market data

Prepared	2026-08-12
Study	Strategy Lab - six parallel analyst streams plus adversarial QA
Primary tests	150 (corrected within-stream and study-wide)
Survivors	0 of 150 in the main study; 1 of 4 in a pooled follow-up (see section 6)
Instruments	56 resolved, 0 refused
Engine	Application's own backtest workbench, rule DSL, and lookahead guards

This document is research, not investment advice. It reports what specific, pre-registered technical conditions did historically, with sample sizes and multiple-testing corrections attached, and characterises instruments by their measured behaviour. It does not recommend any security, does not propose entry or exit prices for anyone to act on, contains no forecast, and is not suitable as the basis for a transaction. Nothing here was prepared by a licensed investment adviser. Historical behaviour is not predictive; every result below is drawn from a single historical path and is subject to the limitations set out in section 9.

1. What was done, and what came out

A 12-page chart-analysis note was treated as a set of falsifiable claims rather than a specification to implement. Seventeen distinct claims were extracted, encoded in the application's rule DSL where expressible, and tested on five instruments chosen so that a large-cap-technology effect could not disguise itself as broad replication (SPY, QQQ, IWM, XLK, XLV).

Each analyst declared a complete family of hypotheses before running anything, judged each idea at one pre-declared forward horizon, and corrected across the whole family with Benjamini-Hochberg. Every result was then pooled into a single study-wide ledger, because five analysts each running about forty tests have collectively run about two hundred, and a q-value inside a forty-test family overstates its case once that is known.

Measure	Result
Primary tests, all streams pooled	150
Survivors, within-stream correction	0
Survivors, study-wide correction	0
Lookahead guards passed	150 of 150 (zero unverified, zero failed)
Strategies beating buy-and-hold on total return	about 1 in 150, and that one underpowered
Seasonality family	75 cells, 0 survivors (all q >= 0.98)
Volatility-threshold family	15 cells, 0 survivors

The smallest p-value anywhere in the pooled primary family was 0.0075, on the note's own headline entry rule, which corrected to q = 0.276 within its stream and q = 0.529 study-wide. A prior study on this codebase found zero survivors in 75 tests; this one found zero in 150 across a wider hypothesis set. Two independent studies, 225 corrected tests, no survivor.

That is a substantive result rather than a failed study. It says these particular rules, on these instruments, at daily resolution, did not produce an edge distinguishable from chance once the number of things tried was accounted for.

Ten smallest p-values in the pooled family

Stream	Symbol	Idea	h	Edge	n	p	q stream	q study
mean-reversion	SPY	headline_conjunction	20	+0.133	87	0.0075	0.276	0.529
mean-reversion	XLV	zscore_extreme_stopout	20	+0.239	21	0.0138	0.276	0.529
momentum	IWM	macd_zero_cross	10	+0.125	98	0.0143	0.391	0.529
momentum	XLV	rsi_oversold_reversion	5	+0.145	65	0.0174	0.391	0.529
trend-ma	XLV	intermediate_cross_ema20_sma50	20	+0.138	74	0.0176	0.793	0.529
momentum	IWM	ma_crossover_matched	10	+0.090	127	0.0394	0.591	0.873
trend-ma	SPY	golden_cross_sma	60	+0.239	15	0.0493	0.793	0.873
mean-reversion	IWM	headline_conjunction	20	+0.120	63	0.0539	0.556	0.873
mean-reversion	SPY	bb_lower_only	20	+0.075	145	0.0697	0.556	0.873
mean-reversion	SPY	zscore_extreme_stopout	20	+0.165	29	0.0843	0.556	0.873

Edge is the trigger's forward hit rate minus the same-bars baseline. n is independent (non-overlapping) windows, not raw triggers. No row clears q <= 0.10.

2. How to read these numbers

Baseline, not zero

A rule that is right 60% of the time is worthless if the instrument rose 60% of the time anyway. Every hit rate here is reported against a baseline computed over exactly the bars that rule could see, so the comparison is like for like.

Independent sample size

Overlapping forward windows are not independent observations. A condition firing on twenty consecutive days at a 20-day horizon is one event observed twenty times. The n column counts non-overlapping windows only; below 30 a result is marked underpowered and not interpreted.

Correction for multiplicity

Testing many rules guarantees some will look good. Benjamini-Hochberg adjusts each p-value for how many tests shared its ledger. The gap between a raw p and a corrected q is where most published technical-analysis results quietly live.

Direction checked before significance

An effect running opposite to the pre-declared direction is a rejection of the hypothesis as written, however large it is, and is reported as rejected rather than inconclusive.

Buy-and-hold on the identical window

Never optional. A strategy returning 300% over 25 years while the instrument returned 1,300% is a losing strategy, and is described that way here.

Frictions are reported, never assumed away

All backtests charge 5 basis points of slippage each way, and cost drag is measured by re-running the whole simulation at zero cost rather than by subtracting an estimate. Several rules in this study lost more to friction than they earned in total.

Lookahead proved, not asserted

Every expression is recomputed on truncated leading slices of history and required to produce bit-identical values before the cut. Anything unproven is treated as a failure.

3. The source note, claim by claim

Seventeen claims extracted from the note. This is the core of the study.

#	Note	Claim	Verdict
1	p.2	IF price > 200MA AND price < lower BB THEN Buy	Best result in the study, still not significant. SPY edge +13.3pp, n=87, p=0.0075; q=0.276 in-strea killed by correction rather than by the data.
2	p.2	200MA as market-regime gate ('trade in a rising market')	Real, but as risk control, not signal. Cuts max drawdown up to half and exposure 15-19pp; makes XLV. Open 67-75% of the time, so a default state rather than a rare permission.
3	p.4	Ideal Entry setup: 20EMA > 50SMA > 10EMA > 200MA	The note's literal ordering beat the conventional stacked reading: positive edge on all 5 symbols, rejections for the 'corrected' version. Neither significant.
4	p.3	'Price crossing MA' differs from 'MA crossing MA'	Qualitatively supported. MA-crossing-MA fires about half as often, with better win rate and far less
5	p.4	Golden cross, 50/200 SMA	Underpowered on 4 of 5 symbols, rejected on the fifth. The event is too rare to test per-instrumen across instruments might.
6	p.6	MACD crossover is faster but more falsy than MA crossover	Split, and the note conflates two signals. Signal-line cross is decisively faster (10-11/yr vs 5.2/yr; cross is slower than the MA crossover. 'More falsy' unsupported: MACD win rate 41.96% vs 39.80
7	p.6	Bullish/bearish divergence	Now testable: a swing-pivot operator was built for this study. Bullish divergence 0 of 45, and it doe Bearish MACD divergence survived a pooled ledger with heavy caveats (section 6).
8	p.5	Z-score band 0.5<|z|<2, hard stop at |z|>3	Underpowered everywhere (n=12-29, all under the 30 floor). The DSL has no abs(), so only the o could not be tested as a mechanism at all.
9	p.5	BB bandwidth <5% compression, >15% high-volatility regime	No survivor. A direction-agnostic claim tested with a directional statistic; median forward return re
10	p.9	Bollinger band touch as an exit	Tested in that role. No survivor.
11	pp.6,8,9	Confluence / 'Golden Alignment' stacking	Costs certain, benefits not. Stacking loses 26-32% of independent sample versus the trigger alon three on 3 of 10 cells, worst on 3 of 10, no pattern.
12	pp.2,7	RSI as oversold trigger and as trend-strength regime	Neither framing survives. Reversion positive on all 5 but best q=0.391 and a net loser on XLK/XL sign across symbols.
13	p.10	March-April bullish, May-October bearish	Does not hold. All 75 cells q >= 0.98, directionally inconsistent on both halves, composites underp largest raw effects were both folklore and both evaporated after correction.
14	p.11	Fear & Greed / VIX-based strategy	VIX partially, Fear & Greed not at all. VIX>30 positive on all 5 symbols (+5.9 to +9.9pp) with consi near-zero or wrong-way on 4 of 5. Fear & Greed has no persisted history, so it is architecturally u
15	pp.2,7	ATR position sizing; max-drawdown cap	Not testable through the engine, which is long-only and all-in with no sizing concept. Descriptive s ATR-proportional sizing cut volatility and drawdown on all 5 symbols and improved Sharpe on 4. T study, and unreachable by the backtester.
16	pp.2,7,10	Volume Profile, ADX, Stochastic, Ichimoku, Fibonacci, order flow	Not in the indicator registry. Not tested. Volume indicators remain deliberately deferred.
17	p.9	Multi-timeframe confluence	Not expressible: one expression sees one interval. Also blocked by the settled daily-bars-only dec

Zero of seventeen survive statistical correction. Four are qualitatively supported as descriptions of mechanics while failing as edges. Five could not be tested because the application cannot express them.

4. Strategy archetype profiles

Every rule tested, with its exact entry and exit expressions and its measured mechanics. Figures are the median across the five primary instruments over full available history, at 5bps slippage each way. 'Beat B&H;' counts how many of the five instruments the rule beat buy-and-hold on total return.

Trend and moving-average framework

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
ideal entry verbatim	`IN ema(20) > sma(50) and sma(50) > ema(10) and ema(10) > sma(200)` `OUT ema(5) crosses_below ema(20)`	2.43	10.48	-29.0%	28.5%	-19.4%	0/5
stacked alignment conventional	`IN ema(10) > ema(20) and ema(20) > sma(50) and sma(50) > sma(200)` `OUT ema(5) crosses_below ema(20)`	2.08	10.48	-32.3%	51.0%	-54.0%	0/5
stacked alignment no gate	`IN ema(10) > ema(20) and ema(20) > sma(50)` `OUT ema(5) crosses_below ema(20)`	2.14	10.48	-54.7%	65.5%	-69.1%	0/5
golden cross sma	`IN sma(50) crosses_above sma(200)` `OUT sma(50) crosses_below sma(200)`	8.26	10.48	-34.1%	65.6%	-14.5%	1/5
intermediate cross ema20 sma50	`IN ema(20) crosses_above sma(50)` `OUT ema(20) crosses_below sma(50)`	2.48	10.48	-43.0%	63.9%	-33.8%	0/5
price cross sma50	`IN close crosses_above sma(50)` `OUT close crosses_below sma(50)`	1.91	10.48	-41.9%	67.0%	-104.0%	0/5
short term cross 10ema 20sma	`IN ema(10) crosses_above sma(20)` `OUT ema(10) crosses_below sma(20)`	2.70	10.48	-48.8%	62.4%	-85.8%	0/5
ma strategy page	`IN close > sma(200) and close > sma(120) and sma(5) crosses_above sma(20)` `OUT close >= bb_upper(20,2) or sma(5) crosses_below sma(20)`	0.21	10.48	-26.1%	17.8%	-20.2%	0/5
regime filter standalone	`IN close > sma(200)` `OUT close < sma(200)`	6.95	10.48	-52.8%	71.0%	-71.8%	0/5

Mean reversion: Bollinger bands, z-score, volatility regime

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
bb lower only	`IN close < bb_lower(20,2)` `OUT close > bb_mid(20,2) [cap 40b]`	1.21	10.48	-43.6%	22.4%	-27.7%	0/5
sma200 only	`IN close > sma(200)` `OUT close < sma(200)`	6.95	10.48	-52.8%	71.0%	-71.8%	0/5
headline conjunction	`IN close > sma(200) and close < bb_lower(20,2)` `OUT close > bb_upper(20,2) [cap 40b]`	2.98	10.48	-28.5%	25.7%	-23.2%	0/5
zscore oversold band	`IN (close - bb_mid(20,2)) / ((bb_upper(20,2) - bb_mid(20,2)) / 2) < -0.5 and (close - bb_mid(20,2)) / ((bb_upper(20,2) - bb_mid(20,2)) / 2) > -2` `OUT close > bb_mid(20,2) [cap 20b]`	1.31	10.48	-46.4%	38.9%	-83.3%	0/5
zscore extreme stopout	`IN (close - bb_mid(20,2)) / ((bb_upper(20,2) - bb_mid(20,2)) / 2) < -3` `OUT close > bb_mid(20,2) [cap 20b]`	0.31	10.48	-17.1%	3.8%	-2.2%	0/5

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
bb bandwidth compression	`IN (bb_upper(20,2) - bb_lower(20,2)) / bb_mid(20,2) < 0.05` `OUT horizon cap only [cap 20b]`	1.17	10.48	-40.7%	40.5%	-39.7%	0/5
bb bandwidth expansion	`IN (bb_upper(20,2) - bb_lower(20,2)) / bb_mid(20,2) > 0.15` `OUT horizon cap only [cap 20b]`	0.74	10.48	-39.9%	16.3%	-3.7%	0/5
boll50 band touch	`IN close < bb_lower(50,2)` `OUT close > bb_mid(50,2) [cap 40b]`	0.47	10.48	-45.9%	19.4%	-8.4%	0/5

Momentum oscillators: MACD and RSI

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
macd signal cross	`IN macd_macd(12,26,9) crosses_above macd_signal(12,26,9)` `OUT macd_macd(12,26,9) crosses_below macd_signal(12,26,9)`	0.53	10.48	-49.3%	54.6%	-48.5%	0/5
macd zero cross	`IN macd_macd(12,26,9) crosses_above 0` `OUT macd_macd(12,26,9) crosses_below 0`	1.96	10.48	-34.7%	65.3%	-34.9%	0/5
ma crossover matched	`IN ema(10) crosses_above ema(20)` `OUT ema(10) crosses_below ema(20)`	2.35	10.48	-42.5%	64.5%	-63.8%	0/5
rsi oversold reversion	`IN rsi(14) < 30` `OUT rsi(14) > 50 [cap 10b, stop 15%]`	0.08	10.48	-42.2%	6.8%	-6.4%	0/5
rsi regime gate	`IN rsi(14) crosses_above 60` `OUT rsi(14) crosses_below 40`	1.62	10.48	-51.2%	61.9%	-21.8%	0/5
trend gate sma200	`IN close > sma(200)` `OUT close < sma(200)`	6.95	10.48	-52.8%	71.0%	-71.8%	0/5
confluence macd trend	`IN macd_macd(12,26,9) crosses_above macd_signal(12,26,9) and close > sma(200)` `OUT macd_macd(12,26,9) crosses_below macd_signal(12,26,9)`	0.22	10.48	-34.0%	35.3%	-24.0%	0/5
trend gate ema20 sma50	`IN ema(20) > sma(50)` `OUT ema(20) crosses_below sma(50)`	2.39	10.48	-43.3%	64.0%	-33.2%	0/5
confluence rsi trend	`IN rsi(14) crosses_above 60 and ema(20) > sma(50)` `OUT rsi(14) crosses_below 40`	1.11	10.48	-39.0%	52.7%	-13.4%	0/5

Regime and risk

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
below 200ma regime	`IN close < sma(200)` `OUT close > sma(200)`	0.62	10.36	-64.5%	28.7%	-19.0%	0/5
regime filtered pullback	`IN close > sma(200) and rsi(14) < 30` `OUT rsi(14) > 50 [cap 15b, stop 10%]`	0.18	10.36	-16.2%	1.6%	-1.2%	0/5
unfiltered pullback	`IN rsi(14) < 30` `OUT rsi(14) > 50 [cap 15b, stop 10%]`	0.50	10.36	-32.3%	8.1%	-7.5%	0/5

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
unfavorable regime pullback	`IN close < sma(200) and rsi(14) < 30` `OUT rsi(14) > 50 [cap 15b, stop 10%]`	0.42	10.36	-32.3%	6.6%	-5.6%	0/5

Divergence (swing-pivot operator)

Rule	Entry / Exit	Med. return	vs B&H	Max DD	Exposure	Cost drag	Beat B&H
bullish rsi divergence	`IN low < swing_low(5) and rsi(14) > rsi_at_swing_low(14,5)` `OUT horizon cap only [cap 10b, stop 15%]`	-0.27	10.36	-49.7%	14.3%	-8.4%	0/5
bullish macd divergence	`IN low < swing_low(5) and macd_macd(12,26,9) > macd_at_swing_low(12,26,9,5)` `OUT horizon cap only [cap 10b, stop 15%]`	0.00	10.36	-31.9%	8.1%	-4.7%	0/5
new low alone	`IN low < swing_low(5)` `OUT horizon cap only [cap 10b, stop 15%]`	0.15	10.36	-54.1%	31.3%	-23.9%	0/5
rsi above swing alone	`IN rsi(14) > rsi_at_swing_low(14,5)` `OUT horizon cap only [cap 10b, stop 15%]`	0.37	10.36	-73.4%	89.5%	-103.7%	0/5
macd above swing alone	`IN macd_macd(12,26,9) > macd_at_swing_low(12,26,9,5)` `OUT horizon cap only [cap 10b, stop 15%]`	1.09	10.36	-65.2%	76.9%	-140.6%	0/5
bullish rsi divergence lb10	`IN low < swing_low(10) and rsi(14) > rsi_at_swing_low(14,10)` `OUT horizon cap only [cap 10b, stop 15%]`	-0.21	10.36	-55.1%	12.8%	-6.6%	0/5
bullish rsi divergence trend gated	`IN low < swing_low(5) and rsi(14) > rsi_at_swing_low(14,5) and close > sma(200)` `OUT horizon cap only [cap 10b, stop 15%]`	0.03	10.36	-25.6%	6.0%	-3.7%	0/5

Total return is a multiple, not a percentage: 2.43 means the equity curve ended 3.43x its start. Cost drag is the total-return difference between the costed run and the identical signal sequence at zero cost, measured by re-simulation.

5. Behaviour by market regime

Descriptive strategy mechanics only, with indicators seeded before each window opens. No p-values: regime windows sit outside every correction family in this study by design.

Window	Cells that never traded	Cells that traded	Of those, profitable
dotcom 2000 2002	8	142	25
bull 2003 2007	2	148	-
gfc 2007 2009	18	132	14
qe bull 2009 2019	0	150	-
covid 2020 2021	2	148	-
bear 2022	28	122	28
recent 2023 now	1	149	-

The crisis windows invert the study's headline, and the inversion needs care. In 2007-09 every cell that traded beat buy-and-hold, because buy-and-hold lost 32-48% and these rules mostly lost less. Only 14 of 132 actually made money. In 2022, 44 cells lost more than buy-and-hold did. 'Beat buy-and-hold' in a crash means lost less, and sometimes it does not even mean that.

A further caution: 18 cells in 2007-09 and 28 in 2022 never opened a position at all. A rule that sits out a bear market has avoided it, which is a real property, but it is loss avoidance rather than edge, and it comes with the cost of sitting out recoveries too.

6. Divergence, and the study's only survivor

The note's divergence claim was untestable in the earlier study: the rule language had no way to locate a prior turning point. An operator was built for this report (confirmed swing pivots, published only after the bars needed to confirm them have formed, with oscillator values sampled at the price pivot). All expressions passed the lookahead guard.

Bullish divergence versus its own controls

Idea	SPY	QQQ	IWM	XLK	XLV
bullish rsi divergence	-0.013	-0.017	+0.016	-0.037	-0.020
bullish macd divergence	-0.036	-0.017	-0.019	-0.047	+0.040
new low alone	-0.000	-0.016	-0.002	-0.024	+0.027
rsi above swing alone	-0.003	+0.000	-0.005	+0.002	-0.006

Edge versus baseline at horizon 5. Divergence does not beat the bare 'new low' control; on SPY and XLV it is worse. Adding the oscillator condition bought nothing.

A correction to the earlier study

An earlier descriptive diagnostic, computed outside the guarded rule engine, reported fewer than one divergence event per instrument per year and 10-day forward returns of +3.9% to +6.4% against a +0.3-0.5% baseline. That is what made divergence the most promising untested idea in the prior report. Run through the real operator, events occur 8.9-11.9 times per year and the forward effect is +0.45% to +0.92% against a +0.29-0.43% baseline: roughly ten times more frequent and seven times smaller. The promising thread was an artifact of an unguarded implementation.

The one survivor, and why it is reported as a lead

Idea	Hit	Baseline	Edge	n	p	q	Verdict
bullish rsi divergence	0.5595	0.5428	+0.0167	1614	0.1855	0.2059	inconclusive
bullish macd divergence	0.5696	0.5424	+0.0272	776	0.1303	0.2059	inconclusive
bearish rsi divergence	0.5340	0.5455	-0.0114	3088	0.2059	0.2059	inconclusive
bearish macd divergence	0.5152	0.5464	-0.0312	2108	0.0042	0.0166	survived

Bearish MACD divergence is followed by a 51.5% five-day up-rate against a 54.6% baseline, in the pre-declared direction, at q = 0.017. It is the only association in 195 corrected tests across two studies to survive its ledger. Four reasons to hold it loosely:

- It fails in the per-symbol family. The same rule on the five primary instruments gave edges of +0.003, +0.002, -0.037, -0.031, -0.006: inconclusive or rejected on all five. A result appearing only after pooling is the characteristic shape of a pooling artifact.
- The pooled n is inflated. 2,108 'independent' windows sums each instrument's non-overlapping windows across 13 instruments that trade the same calendar days and overlap by sector. The true p is larger than 0.0042 by an unknown factor.
- It is not tradeable through this engine. It is a downward-direction hypothesis and the backtester is long-only. Nothing here demonstrates a profitable strategy.

- The effect is small: 3.1 percentage points on a 54.6% base rate, before costs, on a signal firing often enough to accumulate meaningful friction.

7. Instrument characterisation

Measured properties of 56 candidates. These are characterisation tables, not selections, and the ranking criterion is stated for each. Nothing here is a recommendation.

Long-horizon character: what an investor would have had to sit through

Symbol	History from	CAGR full	CAGR 10y	Vol	Sharpe	Max drawdown	Worst episode	Days to recover	% c
BIL	2007-05-30	-0.0%	0.0%	0.8%	-0.00	-1.4%	2008-09-29 to 2023-12-18	not recovered	
SHY	2002-07-30	0.0%	-0.4%	1.6%	0.03	-7.2%	2020-04-30 to 2023-03-08	not recovered	
AGG	2003-09-29	-0.2%	-1.4%	5.3%	-0.02	-23.4%	2020-08-04 to 2023-10-19	not recovered	
BND	2007-04-10	-0.2%	-1.5%	5.3%	-0.01	-24.0%	2020-08-06 to 2023-10-19	not recovered	
IEF	2002-07-30	0.5%	-1.9%	6.9%	0.11	-27.7%	2020-08-04 to 2023-10-19	not recovered	
USMV	2011-10-20	9.7%	8.1%	13.5%	0.76	-33.1%	2020-02-14 to 2020-03-23	371	
SCHD	2011-10-20	10.0%	9.2%	15.5%	0.69	-33.4%	2020-01-23 to 2020-03-23	231	
QUAL	2013-07-18	12.1%	12.7%	17.1%	0.75	-34.1%	2020-02-19 to 2020-03-23	154	
MTUM	2013-04-18	14.5%	14.8%	20.2%	0.77	-34.1%	2020-02-19 to 2020-03-23	107	
VOO	2010-09-09	13.0%	13.5%	17.0%	0.80	-34.3%	2020-02-19 to 2020-03-23	148	
QQQM	2020-10-13	16.6%	16.6%	22.3%	0.80	-35.6%	2021-11-19 to 2022-12-28	350	
SPLV	2011-05-05	7.5%	6.0%	14.5%	0.57	-36.6%	2020-02-14 to 2020-03-23	471	
XLP	1998-12-22	4.3%	4.5%	15.3%	0.35	-37.8%	2000-12-28 to 2003-03-10	1724	
XLRE	2015-10-08	3.6%	2.8%	20.4%	0.27	-39.3%	2020-02-21 to 2020-03-23	393	
VLUE	2013-04-18	11.0%	11.8%	18.8%	0.65	-39.5%	2020-02-12 to 2020-03-23	295	
XLV	1998-12-22	7.1%	8.5%	17.9%	0.47	-40.6%	2007-12-10 to 2009-03-09	1100	
GLD	2004-11-18	10.7%	12.1%	18.2%	0.65	-45.6%	2011-08-22 to 2015-12-17	1686	
XLC	2018-06-19	10.2%	10.2%	22.2%	0.55	-47.2%	2021-09-01 to 2022-11-03	602	

Ordered by shallowest maximum drawdown, restricted to instruments with at least 1,000 daily bars. ***Read this with the history-length caveat:*** *funds launched after 2010 have never been tested against 2000-02 or 2008-09 and rank well partly for that reason. QQQ, the same index family as several top-ranked names but with longer history, ranks 43rd of 52.*

Short-horizon (swing) character

Symbol	ATR% (3y)	ATR% (full)	20d vol (3y)	Avg abs daily move	1d autocorrelation	Variance ratio 5d	Variance ratio 20d
SOXL	9.02%	7.13%	107.6%	5.24%	-0.070	0.789	1.000
TQQQ	4.72%	4.70%	56.1%	2.74%	-0.044	0.820	0.913
TSLA	4.70%	4.57%	55.4%	2.66%	-0.008	0.903	1.051
NVDA	3.80%	4.57%	44.2%	2.17%	-0.082	0.829	0.773
UPRO	3.41%	3.73%	40.5%	1.96%	-0.024	0.822	0.855
ARKK	3.28%	2.95%	37.9%	1.91%	-0.042	0.875	0.816
QLD	3.14%	3.36%	37.5%	1.84%	-0.051	0.805	0.875
SOXX	2.93%	2.55%	36.2%	1.78%	-0.098	0.742	0.891
META	2.91%	2.94%	34.5%	1.59%	-0.036	0.792	0.919
SLV	2.84%	2.41%	37.0%	1.79%	-0.012	0.821	0.609
SMH	2.80%	2.74%	34.4%	1.69%	-0.084	0.753	0.844
AMZN	2.60%	3.97%	30.0%	1.45%	0.026	0.879	0.821
GOOGL	2.52%	2.38%	29.1%	1.39%	0.017	0.955	0.913
XBI	2.48%	2.60%	26.9%	1.35%	-0.052	0.801	0.738
SSO	2.28%	2.72%	27.0%	1.30%	-0.031	0.802	0.821
AAPL	2.17%	3.39%	24.4%	1.15%	0.052	1.045	1.023
IGV	2.12%	2.01%	24.4%	1.21%	0.071	1.042	1.073
MSFT	2.04%	2.62%	23.4%	1.11%	0.019	1.124	1.023

Ordered by trailing-3-year average true range as a percentage of price. A variance ratio near 1.0 is random-walk behaviour; below 1.0 is mean-reverting, above 1.0 trending. Negative 1-day autocorrelation indicates day-to-day reversal. Daily range matters practically because a rule whose average move is smaller than the round-trip trading cost cannot profit regardless of how often it is right.

Near-duplicate pairs: where a diversified-looking list is one position

Pair	Correlation	Pair	Correlation	Pair	Correlation
QQQ / QQQM	1.000	QLD / TQQQ	1.000	SSO / UPRO	1.000
QQQM / QLD	1.000	QQQ / QLD	0.999	QQQM / TQQQ	0.999
QQQ / TQQQ	0.999	VOO / SSO	0.999	VOO / UPRO	0.998
SOXX / SOXL	0.997	SPY / SSO	0.997	SPY / VOO	0.997
SPY / UPRO	0.996	VOO / VTI	0.995	VTI / SSO	0.995
SPY / VTI	0.995	VTI / UPRO	0.994	BND / AGG	0.993

Pair	Correlation	Pair	Correlation	Pair	Correlation
SOXX / SMH	0.981	SMH / SOXL	0.980	IEF / BND	0.979
IEF / AGG	0.978	VEA / EFA	0.970	QUAL / SSO	0.970

80 pairs exceed 0.90 correlation over the trailing three years. Broad-market and quality/momentum factor funds cluster tightly with each other and with leveraged products on the same underlying.

Leveraged products: the linear intuition is wrong in both directions

Product	k	Window	Realised	k x underlying	Gap
QLD	2.0x	trailing 1y	44.6%	50.8%	-6.2%
QLD	2.0x	trailing 3y	184.1%	188.0%	-3.8%
QLD	2.0x	trailing 5y	136.3%	191.0%	-54.7%
QLD	2.0x	since inception	7959.1%	3514.8%	4444.3%
TQQQ	3.0x	trailing 1y	60.0%	76.3%	-16.3%
TQQQ	3.0x	trailing 3y	261.8%	282.0%	-20.2%
TQQQ	3.0x	trailing 5y	115.2%	286.4%	-171.2%
TQQQ	3.0x	since inception	33680.9%	4635.5%	29045.4%
SOXL	3.0x	trailing 1y	407.6%	355.2%	52.5%
SOXL	3.0x	trailing 3y	466.9%	651.5%	-184.6%
SOXL	3.0x	trailing 5y	205.0%	753.3%	-548.3%
SOXL	3.0x	since inception	19582.1%	9576.7%	10005.4%
SSO	2.0x	trailing 1y	38.6%	42.3%	-3.7%
SSO	2.0x	trailing 3y	141.5%	143.9%	-2.4%
SSO	2.0x	trailing 5y	121.4%	146.2%	-24.8%
SSO	2.0x	since inception	1492.4%	1032.8%	459.6%
UPRO	3.0x	trailing 1y	56.4%	63.5%	-7.1%
UPRO	3.0x	trailing 3y	219.1%	215.9%	3.2%
UPRO	3.0x	trailing 5y	142.2%	219.4%	-77.1%
UPRO	3.0x	since inception	12621.1%	2210.5%	10410.6%

Gap is realised return minus k times the underlying's return. It is negative in choppy trailing windows (decay) and strongly positive since inception (convex compounding through sustained trends). In worst-episode comparisons the naive k-times-underlying figure falls below -100%, a mathematically impossible fund return, which is the clearest demonstration that the linear model is simply the wrong model rather than merely an optimistic one.

8. Conditions at the time of writing, and study integrity

Market conditions

Measured 2026-08-11, one session before this report's date, and therefore **predating the CPI release scheduled for 2026-08-12**. Treat rate-sensitive figures as stale.

Breadth	12 of 14 tracked instruments above their own 200-day SMA; only XLU and XLC below
Index level	SPY and IWM within 0.35% of 52-week highs
Volatility	VIX 15.28: 13th percentile trailing 1 year, 34th percentile since 1990
Rates	10-year approximately 4.65-4.70%; 2s10s +47bp, positive rather than inverted
Policy	Target 3.50-3.75%, on hold since a 9-3 vote on 2026-07-29 with three hawkish dissents
Dispersion	3-month sector spread 21.45pp (XLV +17.98%, XLC -3.48%)
Unmatched condition	An oil-supply disruption running since 2026-02-28 affecting roughly 20-25% of seaborne trade, with core PCE at 3.3% year on year. No window in this study's sample contains a comparable shock.

Defects found during the study

Recorded because a study that reports no problems with itself has usually not looked.

Where	What
Application	Concurrent readers of the bar cache collide, and the failure surfaces as a per-symbol data refusal indistinguishable from a nonexistent ticker. It cost four instruments before being caught; all four resolved on serial retry.
Application	The market-data staleness gate left most symbols four days stale, which is benign for long backtests and not benign for a live macro panel.
Application	The leveraged-instrument registry has no entry for two products, including the one named in the source note, so its decay guard never sees them.
Application	Fear & Greed has no persisted history, making one strategy family in the note untestable rather than merely unsupported.
Study	The regime-window list initially omitted the 2007-09 crisis entirely. Found by an analyst, not by the author of the brief. Backfilled.
Study	The regime-window slicer recomputed indicators from the start of each window, so a 200-day-average rule inside a 253-bar window was blind for 200 of them. Found by the QA reviewer. It inflated a headline count from 18 to 41 before correction. No p-value was affected, because regime windows carry none.

9. What this report cannot tell you

- **One historical path.** Every independent n counts non-overlapping windows of a single sequence of events, not draws from different economic universes. Eighty-seven independent observations of one 27-year history is not eighty-seven independent experiments.
- **Correlated instruments.** SPY, QQQ and XLK are three views of the same large-cap growth complex. IWM and XLV were included specifically so this would be visible rather than hidden, and several rules did behave differently on them.
- **Survivorship, twice over.** Exchange-traded funds hold today's constituents, and the long-horizon table is dominated by funds too young to have experienced the worst drawdowns in the sample.
- **No fees, spreads, or taxes.** Five basis points of slippage is modelled; expense ratios and real bid-ask spreads are not. For a short-horizon rule a wide spread can exceed the entire modelled effect.
- **Position sizing was never testable.** The most encouraging risk result in the study sits outside the guarded engine and carries no statistical claim whatsoever.
- **Daily bars only.** Intraday history is capped at 60 days by the data provider, so every intraday and multi-timeframe claim in the source note is out of scope rather than refuted.
- **A backtest is not a forecast.** Zero survivors describes what these rules did on stored history under pre-registration, correction and guard discipline. It is not a statement about what any rule will do next, and the current oil-supply condition has no analogue in the sample at all.
- **This is not advice.** No instrument in this document is presented as one to buy or sell, no entry or exit price is proposed for anyone to act on, and no part of it was prepared by a licensed investment adviser. Any decision, and its consequences, remain the reader's own.

Sources

All figures are generated at build time from the study's JSON outputs under docs/research/strategy-lab/data/. Per-stream reports, the scripts that produced every number, and the adversarial QA review are in the same directory tree. This document is reproducible by re-running meta/build_report_pdf.py.